

Full Stack Web Development

Past Year Questions (PYQs) Mapped to Syllabus
(Strictly Organized by Unit on Separate Pages)

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Unit 1: HTML and CSS

HTML

- What is an image map in HTML? Take an Indian map as an image and make three states clickable:
 - Uttarakhand (shape-circle, onclick of the state `https://uk.gov.in/` should open, title displayed on the state-UK)
 - Uttar Pradesh (shape-polygon, onclick of the state `https://up.gov.in/en` should open, title displayed on the state-UP)
 - Rajasthan (shape-rectangle, onclick of the state `https://rajasthan.gov.in/` should open, title displayed on the state-RJ)
- Divide the browser window using **FRAMES**.
- Write short notes on **textarea**, **placeholder** attribute, radio buttons, and checkboxes with suitable examples.
- Create a nested list as given in the diagram (Computer System → Input Devices → Keyboard, Mouse, etc., and Output Devices).
- Create a structure using HTML tables holding grid values (A, B, C, D, E, F, G, H, I).
- What are semantic tags in HTML5? Explain any 5 semantic tags.
- Write short notes on the following tags: **iframe**, **textarea**, **meta**, and **radio** with suitable examples.
- Explain the role of the browser in rendering an HTML document. How do you add meta tags to provide information about a web page?
- Create a static web page using the **Table** Tag of HTML (create a specific Specification Table with Hours and Marks).
- Explain what HTTP Protocol is, and what HTTP Request and HTTP Response are with a diagram.
- Explain the history of WWW. Also, explain how WWW is different from the Internet.
- Discuss the concepts of effective web design in detail.
- What are the key features introduced in HTML5 compared to previous versions? Explain the purpose and usage of semantic HTML5 elements such as **<header>**, **<footer>**, **<nav>**, and **<article>**.
- Define the World Wide Web (WWW) and explain its role in modern internet usage. Discuss the key components of the World Wide Web architecture, including web browsers, web servers, and URLs.
- What is the importance of **method**, **action**, and **enctype** attributes in an HTML form?
- Describe the concept of why HTTP is a stateless protocol.

- Design and write the code for an HTML5 Form incorporating: Slider, Numeric spinner, Date picker, Color picker, Text field with placeholder, URL field, and Email field.
- What is the difference between DIV and SPAN tags? List various HTML tags and use them to develop user-friendly webpages.
- HTML-4 and HTML-5 have some differences. Can you state a few differences along with the explanation of audio and video tags of HTML-5 with its various attributes?
- Explain some differences between HTML-4 and HTML-5. Can you state a few differences along with the explanation of audio and video tags of HTML-5 with its various attributes? Write down a JavaScript code that controls the play and pause of audio and video on a page using a toggle button.
- What is the importance of cross-browser compatibility in web design? Discuss how limited bandwidth affects multimedia-heavy websites and how to optimize for it.
- Explain the difference between GET and POST method. Discuss the use of event listeners in JavaScript to capture form submission and manipulate form data (e.g., preventing default form submission, dynamically changing form content).

CSS

- Develop and demonstrate the usage of inline, internal, and external style sheets using CSS: Use different font styles, control the repetition of images with **background-repeat**, define style for links as **a:link**, **a:active**, **a:hover**, **a:visited**, and add customized cursors for links.
- Explain different types of CSS with examples. Write an HTML page that contains a selection box with a list of 5 countries. When the user selects a country, its capital should be printed. Add CSS to style the capital using: text color, bold, font size, italics, and underline.
- What are the different methods available for including CSS in an HTML document? Please describe all the possible ways to incorporate CSS into HTML and discuss the advantages and disadvantages of each method.
- Which HTML tag is used to define an internal style sheet? (Short Answer / Fill in the blank)
- Write a program to create a Box Shadow and a Text Shadow using CSS3. Also, create Rounded Corners using CSS3.
- Create a web page to show a fixed background image (this image will not scroll with the rest of the page).
- Explain all the CSS background properties with the help of suitable examples.
- Explain all five CSS positioning properties with suitable examples.
- Explain the CSS Box model in detail. Also, explain the difference between margin and padding clearly with suitable examples diagrammatically.

- Discuss the CSS box model and its components, including content, padding, border, and margin. How does the box model influence the layout of elements on a web page?
- Explain the concept of CSS selectors and provide examples of different types of selectors, including element selectors, class selectors, and ID selectors.
- Differentiate between Pseudo-classes and Pseudo-elements in CSS with a suitable example.
- What is a selector in CSS? Can you attach more than one declaration to a selector? What is cascading order?
- Define the CSS Box Model and use selectors to provide styles to the web pages at various levels.
- Using an example, explain how you can calculate the element width and height when you are given the width and height of various parts of the CSS box model.
- Use CSS to create a grid of images with a hover effect that enlarges the images and displays a caption.
- Create an HTML table with data about books and use CSS to style the table with alternating row colors and a fixed header row.
- Write an HTML snippet with CSS for following effects: 1. link: after visited red color and before visited blue color. 2. table with odd rows blue and even rows green color. 3. Text color-green, Text case-uppercase.
- Explain CSS Pseudo-classes, Pseudo-entity, and type selectors.

Unit 2: JavaScript and Advance JavaScript

JavaScript

- Develop and demonstrate JavaScript with POP-UP boxes and functions for the following: click on a button to display the date, take a number using `prompt` and output its factorial using `alert`, take a number using `prompt` and output a multiplication table using `alert`, take a number using `prompt` and add another using `confirm` to output their sum.
- Explain the concept of variables in JavaScript. What are the different variable scopes, and how are variables declared?
- Write a JavaScript program to display the current time and date in the following format: **Date: Date/Month/Year, Time: hour/minute/second.**
- Create a JavaScript object called `student` with properties like name, age, grade, and a method to display the details. Demonstrate how to access and modify its properties.
- How can you format a date in DD/MM/YYYY format using the Date object? Create a JavaScript function to display the current date and time.
- Briefly discuss event handling from body elements and button elements in JavaScript.
- Write a program in JavaScript to input a string from a user through a text field and convert the string to Uppercase on the `onfocus()` event and Lowercase on the `onblur()` event.
- Write a JavaScript to display the timer clock on a webpage using methods `setInterval()` and `setTimeout()`.
- Explain how events are handled in JavaScript. Write an event handler program using a function to hide email addresses to protect from unauthorized users.
- Write down a JavaScript code that controls the play and pause of audio and video on a page using a toggle button.
- Write a JavaScript function that takes an array of numbers as an argument and returns the sum of all the even numbers in the array.
- Write code to create an HTML page to explain input and output using a calculator with the use of various predefined functions and objects in JavaScript.
- Write a Program in HTML, CSS and JavaScript to add a calculator on a web page (Simple Calculator with HTML CSS and JavaScript).
- Write a JavaScript function that takes an array of numbers as input and returns a new array with the numbers sorted in descending order.
- Write a JavaScript program to create a button that, when clicked, displays an alert with a random number generated between 1 and 10.
- Explain `setInterval`, `clearInterval`, `setTimeout`, and `clearTimeout` methods in JavaScript.
- True/False: The external JavaScript file must contain the `<script>` tag.

Advance JavaScript (DOM, Objects, Form Validation, JSON)

- Write JavaScript to validate fields of a Registration page: First Name (alphabets only, length > 6), Password (first letter capital, one number, one special char, length > 6), E-mail id (valid pattern), Mobile Number (10 digits), and Address (not empty).
- Explain Document Object Model (DOM) with an example. Write code for a web application which accepts the birth date from the user in a textbox and displays the day of the week in a message box on the click of a button.
- Create an HTML form accepting name, age, and nationality. Using JavaScript, check conditions for casting a vote (age ≥ 18 and nationality is Indian).
- Assume a paragraph containing dummy text. On the click of a button, modify and then show the content of this paragraph using any two JavaScript DOM methods.
- How is form validation done using JavaScript? Write a script to validate an Employee Details form using regular expressions.
- Design a login form using HTML & JavaScript with following validations on username and password fields: 1. Password length must be 6 to 12 characters. 2. Username should not start with @ or number. 3. Phone number must be 10 digit start with 6/7/8/9. 4. All fields should not be blank.
- What does DOM stand for? Explain the topmost objects in the DOM. Explain the Navigator Object in detail.
- Write an HTML page that contains a selection box that contains the names of 5 projects. Only three students are allowed to work on each project. On selecting a project, the list of students working on it should be displayed. Add CSS to style the list using the following properties: text color, bold, font size.
- Write a function in JavaScript to insert a string within a string at a particular position.
- Define JSON and explain its purpose in web development. Describe the syntax of JSON and provide examples of JSON data structures.
- Is JSON better than XML? Explain your answer.
- How do you validate a form input field in JavaScript? Create a form validation script that checks if the username and password fields are not empty.
- Explain how validations are done on the client-side and server-side. Which one is more secure? Justify your answer with a form validation using regular expressions.
- Differentiate between client-side and server-side validation with examples. Which one is more secure, and why? Write a code snippet that validates an email address and mobile number using regular expressions on server side.

Unit 3: XML, Ajax, and JQuery

XML

- Define XML and list the rules to be followed by XML documents. What is the XML Namespace? How to use CSS inside an XML file?
- How is XML used in web applications? Write a sample XML file that stores a list of books (title, author, year) and create a simple DTD or Schema to validate it.
- What is XML and DTD? Explain the relationship between XML and DTD with some suitable examples.
- Define XML and list the rules to be followed by XML documents. What is a PCDATA and CDATA section in DTD? With the help of XML, create a database of Courses and validate it with the help of DTD.
- Define DTD (Document Type Definition). What is a PCDATA and CDATA section in XML? With the help of XML, create a database of employees and validate it with the help of DTD.
- Design an XML document to store information about a student of a Graphic Era University. The information must include Name, Branch, Year of Joining, and e-mail id. Make up sample data for 3 students. Create a CSS style sheet and use it to display the document.
- Write XML file to store information about Music_Album having fields of Title, Singer, Music_Composer, Actor, Language, Country and Release_Year. Also write corresponding external DTD file.
- What is the difference between XML and JSON? Which one is better for storing unstructured databases?

Ajax

- What is Ajax? Write a small program to make an Ajax call to a server and insert the response web page.
- Explain Ajax with an example.
- How does XMLHttpRequest work in AJAX?
- Explain the concept of AJAX? Write a program to fetch the content of a JSON file named json_demo.json with the help of AJAX and visualize the database on the client side.

JQuery

- Describe the purpose of jQuery effects and how they can enhance user interaction on web pages.
- What is the difference between JavaScript and jQuery? What are the two ways of using jQuery on your website and which one of them is better and why?
- Explain the difference between JavaScript and jQuery? What are the two ways of using jQuery on your website, and which one of them is better, and why? Please write the

syntax of jQuery and explain how you can hide and unhide an element (using element tag, class, and ID for selecting element) when the mouse enters or leaves that element.

- Write the syntax of jQuery and explain how you can hide and unhide an element (using element tag, class, and ID) when the mouse enters or leaves that element. Explain why you should use the document ready event.
- Write a program to design a curriculum of a course and traverse it with the help of DOM manipulation using jQuery.
- Explain the concept of AJAX/jQuery. Write a program to fetch the content of a JSON file named `json_demo.json` with the help of AJAX and visualize the database on the client side.

Unit 4: PHP and Advance Features

PHP Basics, HTML, Arrays, Functions, Strings, and Files

- What is PHP and explain the advantages and disadvantages of PHP. Mention 5 reasons why PHP is so popular. What does PHP stand for?
- Write a PHP program to accept values from a form (name, residential address, email, mobile). Apply server-side validations on it and insert the data into an appropriate table or show an error.
- Write a program in HTML (Web-Page) having a form with firstname, lastname and a submit button. On clicking the button form validation should work (Use Javascript for that) and data entry should be saved in a database (PHP code needed). Form should be styled using CSS.
- What is an associative array in PHP and how does it differ from an indexed array? Explain a few functions for sorting in associative arrays in PHP and also write PHP script to create an associative array of student names and their marks. Sort the array by values in ascending order and by keys in descending order.
- Explain the associative arrays in PHP along with the detailed description of `$_GET` and `$_POST` global variables? Also, use a suitable example, to explain the way you can extract the data from `$_GET` and `$_POST` from server side using PHP.
- Explain some of the sorting methods of associative arrays in PHP? Write a program to declare an associative array and sort them in ascending order with the help of keys.
- Explain what is an array and string in PHP. Write a PHP program to accept a user's firstname and lastname, combine them, and output the result.
- Explain the PHP array sort functions with examples. Write a PHP script to sort a given associative array in descending order according to key.
- Create a PHP function that takes an array of numbers as input and returns the sum of all the numbers in the array.
- What is the use of `explode()` and `split()` functions in PHP? Write a program to implement `explode()`, `implode()`, `compact()`, and `extract()`.
- What is meant by an associative array in PHP? Write a PHP script to create an associative array of student names and their grades.
- Write a PHP program for creating and manipulating: Indexed array, Associative array, Multidimensional array.
- Write a PHP program to demonstrate the variable functions `gettype()`, `settype()`, `isset()`, and `unset()`.
- Describe file handling in PHP. Write syntax for file handling like open file, read file, write file in PHP.
- Describe the typical steps involved in reading from and writing to a file using PHP. Write a PHP script that creates a text file named log.txt, writes a message to it, and then reads and displays its contents on the browser.

- Describe file handling in PHP, With the help of PHP code insert some text in text file. After inserting the text read the text from the same text file and shows it on web page.
- Write a PHP program to open a file and read contents char by char till the end of the file, find all the consonants in a separate file and vowels in a separate file. Explain `fopen()` permissions.
- Write client-side and server-side code to save an uploaded file in PHP. Write a program to upload an image to the server (check type is image, size ≤ 50000 , jpeg only).
- Describe the steps involved in securely uploading a file to the server and checking for potential security risks like file size, type, and extension. What is the role of the `$_FILES` superglobal?
- Explain `include` vs `require` function in PHP. How are execution failures handled? What is the main difference between `require()` and `require_once()`?
- What is the difference between `$message` and `$$message`?
- What is the use of the `header()` function in PHP? What are magic constants in PHP?
- Write a PHP script to show global and static scope of a variable.
- Explain the concept of error handling in PHP. How can errors and exceptions be handled effectively?
- PHP server scripts are surrounded by delimiters. (Fill in the blank)

Advance Features (Cookies, Sessions, and Database)

- What are cookies and sessions in PHP? Explain both with the help of a program. Explain 3 differences between a cookie and a session.
- What is the difference between sessions and cookies in PHP? Compare them in terms of storage location, security, and typical use-cases. How data can be stored and retrieved using sessions and cookies in PHP and also explain how session data is destroyed and how cookies can be deleted from the client browser.
- Explain the concept of sessions in PHP. How are sessions maintained, and how can session variables be accessed?
- Write a program to design a login page with remember me. If checked, store credentials in cookies and sessions. Destroy sessions upon logout.
- Write a PHP program to store current date-time in a COOKIE and display the 'Last visited on' date-time on the web page upon reopening of the same page.
- Write a PHP script to implement super global variables `$_GET`, `$_POST`, and `$_REQUEST`. Differentiate between GET and POST methods, explaining use cases and security.
- True/False: `$_GLOBALS` is a Super global variable.
- How to create a MySQL connection and Database in PHP? How do you connect to a MySQL database using PHP?

- Describe how to establish a connection to a MySQL database using PHP with mysqli or PDO. Write PHP scripts to insert, update, and delete records in a students table with the following fields: EnrolmentNo, Username, Name, Email, PhoneNo, Gender, Country.
- Write a PHP program to query a database and check if a table is empty or not. Print rows from the table.
- Write a PHP script to create a student table and insert at least 5 records in it.
- Create a Student database with the following fields using MySQL queries and perform any 5 basic queries on the table. Name, Reg No, Course, College, Grade.
- Write a PHP program to connect to a database and check if a username-password combination is correct. Show a welcome message or redirect to login upon error.
- Create an HTML form taking username and password. On login, redirect to an index page with student records. Write insert, edit, and delete queries on the records table and execute them with PHP.
- How can we retrieve the data in the result set of MySQL using PHP?
- Write a PHP program to upload a registration form into a database and display the registration form from the database.
- What is Cross-Site Scripting (XSS)? Explain the different types of XSS attacks (Stored, Reflected, and DOM-based) with examples. Write a PHP program that includes a form and safely displays submitted user input. Use `htmlspecialchars()` and `$_SERVER['PHP_SELF']` to demonstrate XSS prevention.

Unit 5: MERN Stack and Web Application Deployment

Web Application Deployment and CMS

- What is a CMS? What are its advantages and disadvantages? List any two popular Content Management Systems (CMS).
- What do you mean by CMS? Based on market share, WordPress is the most popular CMS, but can you name a few other CMSs? Please list down some of the features of CMSs and describe WordPress briefly from a technical perspective.
- Analyze the role of CMS in modern web development. Compare WordPress with at least two other popular CMS platforms in terms of features, flexibility, and use-cases. From a technical perspective, explain how WordPress handles themes, plugins, and database interactions.
- What is WordPress, and how does it differ from other content management systems (CMS)?
- What is a sitemap and its use? How to make a website look good on all resolutions?
- List the stages of publishing a website. Discuss tools and techniques involved in each stage.

MERN Stack (MongoDB, Express, React, Node)

- Explain the components of the MERN stack (MongoDB, Express.js, React.js, and Node.js) and how they interact in a typical full-stack web application. Illustrate the request-response lifecycle through the stack with a simple use-case like user login or data submission.
- Explain what React is and its core principles.
- What are the differences between React class components and functional components?
- Describe the purpose of JSX in React. How does JSX help in writing React components?
- Describe the component lifecycle in Class components using React JS.
- How does Axios differ from `fetch()` for making API requests?
- Write a React component that renders a form with name and email fields, and logs the data on submission.
- What is the Virtual DOM and how does it improve performance?
- Compare MongoDB with MySQL in terms of data storage, performance, and scalability.